advance on previous systems, having a storable propellant that it gave it mobility as well as enhanced operational capabilities. Western intelligence believed it had a range of 1,100 miles (even though Oleg Penkovsky had told them it was 1,300 miles). According to Norman Polmar and John Gresham, US intelligence underestimated the range of the Soviet MRBMs and IRBMs by over twenty per cent.[72] In either case, the R-12 could target Washington, among many other US (as well as Central and South American) cities.[73]

The R-14s had only recently been introduced into service and around thirty were then targeted at NATO Europe. They were fired from fixed sites (though not hardened silos) and had a range of over 2,200 miles according to Western intelligence, or 2,800 miles according to Soviet sources.[74] In the event, the IRBMs were turned around to avoid the blockade and never made it to Cuba.

The naval contribution of Operation Anadyr was Operation Kama, which included submarine-based nuclear missiles. The Soviet navy was to construct a naval base at Mariel, from which Project 658 (NATO designation Hotel) nuclear-armed and nuclear-powered submarines (SSBNs) would operate. However, none were available in the summer of 1962 as they were undergoing refits following a near catastrophe on board the first Soviet SSBN, the K-19, in July 1961, when it suffered a reactor leak. In 2006 Mikhail Gorbachev nominated Captain Nikolai Zateev and members of his crew for the Nobel Peace prize for their heroism and self-sacrifice in averting a nuclear explosion. The events on the K-19 were grippingly portrayed in the 2002 film *K-19: The Widowmaker* (starring Harrison Ford and Liam Neeson).

Pending availability of the SSBNs, seven Project 629 submarines (NATO designation Golf) were earmarked for Operation Anadyr, each equipped with three 325-mile-range R-13 missiles (NATO designation SS-N-4 Sark). These carried explosive yields comparable to the MRBMs. The Project 629s were powered by diesel-electric engines and, unlike nuclear-powered submarines, needed to surface to charge their batteries. They were thus more vulnerable to US Anti-Submarine (ASW) ships and aircraft. In addition, while US Polaris boats could fire their solid-fuelled missiles when